Exercise 1: SQL Fundamentals

1. SELECT ~~* FROM~~ Statement

```
SELECT * FROM employees_db;
```

output

id	first_name	last_name	department	salary	hire_date	City
1	John	Doe	IT	55 000	2018-06-15	New York
2	Jane	Smith	HR	48 000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60 000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53 000	2021-03-25	New York
5	David	White	Marketing	52 000	2016-04-10	San Francisco
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59 000	2019-10-01	Houston
8	Jessica	Moore	HR	51 000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53 000	2022-06-01	Chicago
10	Laura	Hall	IT	50 000	2020-08-10	San Francisc

2. SELECT DISTINCT Statement

```
SELECT DISTINCT department
FROM employees_db;
```

OUTPUT

↓

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department
IT
HR
Finance
Marketing

3. ORDER BY Statement

```
SELECT  first_name,
        last_name,
        Salary
FROM    employees_db
ORDER BY salary DESC;
```

OUTPUT

first_name	last_name	Salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Sarah	Brown	53 000
Daniel	Clark	53 000
David	White	52 000
Jessica	Moore	51 000
Laura	Hall	50 000
Jane	Smith	48 000

4 LIMIT statement

```
SELECT *
FROM employees
ORDER BY salary DESC
LIMIT 5;
```

id	first_name	last_name	Salary
1	Emily	Davis	62 000
2	Mike	Johnson	60 000
3	Robert	Wilson	59 000
4	John	Doe	55 000
5	Sarah	Brown	53 000

5 WHERE statement

```
SELECT first_name,
       last_name,
       department
FROM employees_db
WHERE department = 'IT';
```

id	first_name	last_name	department
1	John	Doe	IT
2	Sarah	Brown	IT
3	Emily	Davis	IT
4	Laura	Hall	IT

6. AND Statement

```
SELECT *
FROM employees_db
WHERE department = 'Finance' AND salary > 58 000;
```

	first_name	last_name	department	Salary
	Mike	Johnson	Finance	60 000
	Robert	Wilson	Finance	59 000

7. OR statement

```
SELECT *
FROM employees_db
WHERE department = 'HR' OR department = 'Marketing';
```

id	first_name	last_name	department
1	Jane	Smith	HR
2	David	White	Marketing
3	Jessica	Moore	HR
4	Daniel	Clark	Marketing

8. NOT statement

```
SELECT *
FROM employees_db
WHERE NOT department = 'IT';
```

OUTPUT ↓

6. AND statement

```
SELECT first_name,
       Last_name
FROM employees
WHERE department = 'finance' AND salary > 58 000;
```

OUTPUT

first_name	last_name	department	salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000

7. OR statement

```
SELECT first_name,
       last_name
FROM employees
WHERE department = 'HR' OR department = 'Marketing'
```

OUTPUT ↓

First_name	last_name	department
Jane	Smith	HR
David	White	Marketing
Jessica	Moore	HR
Daniel	Clark	Marketing

8. NOT statement

```
SELECT *
FROM employees
WHERE NOT department = 'IT';
```

OUTPUT

first_name	last_name	department
Jane	Smith	HR
Mike	Johnson	Finance
David	White	Marketing
Robert	Wilson	Finance
Jessica	Moore	HR
Daniel	Clark	Marketing

9. IN statement

```
SELECT *
FROM employee
WHERE department IN ('HR', 'IT', 'Finance');
```

output

first-name	last-name	department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laura	Hall	IT

10. Combining Conditions

```
SELECT *
FROM employee
WHERE department = 'IT' AND salary > 50 000
                AND city = 'New York' ;
```

OUTPUT

First_name	last_name
John	Doe
Sarah	Brown

11. Combining WHERE, AND, and ORDER BY

```
SELECT first_name,
       last_name
FROM employee
WHERE department = 'finance' OR department = 'Marketing'
      salary > 52 000
ORDER BY salary DESC
```

OUTPUT

First_name	last_name
John	Doe
Sarah	Brown

12. Combining SELECT DISTINCT, WHERE, and IN

```
SELECT DISTINCT city
FROM employee
WHERE department NOT IN ('IT','HR');
```

output

first_name	last_name	department
Mike	Johnson	Finance
David	White	Marketing
Robert	Wilson	Finance
Daniel	Clark	Marketing

13 Combining WHERE, NOT, AND, and ORDER BY

```
SELECT *
FROM employees
WHERE NOT department 'finance' AND salary > 50 000
ORDER BY hire_date ASC
```

output

first_name	last_name	salary
John	Doe	
Sarah	Brown	
David	White	
Emily	Davis	
Jessica	Moore	
Daniel	Clark	

first_name	last_name	salary
Jessica	Moore	51 000
David	White	52 000
Sarah	Brown	53 000
Daniel	Clark	53 000
John	Doe	55 000
Emily	Davis	62 000

14. Combining WHERE, OR, IN, and LIMIT

```
SELECT *
FROM employee
WHERE city = 'Chicago' OR city = 'Los Angeles' AND
      department = 'IT' OR department 'Marketing'
LIMIT 3;
```

~~WHERE City IN ('Chicago~~

Output:

first_name	last_name	department	City
Jane	Smith	HR	Chicago
Mike	Johnson	Finance	Los Angeles
Emily	Davis	IT	Chicago